

PRST (14/05/2021)

Convergence in probability

Let $\{X_n\}$ be a sequence of rvs

$\{X_n\}$ converges to the rv X

in probability if

for every $\epsilon > 0$, $\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) = 0$
 or $\lim_{n \rightarrow \infty} P(|X_n - X| < \epsilon) = 1$.

X_1
 X_2
 X_3
 $\vdots$

$$X_n \xrightarrow{P} X$$

X_1 : outcome due to the 1st throw of a die
 X_2 : outcome 2nd u
 X_3 : outcome 3rd
 $\vdots$
 X_n : outcome nth u

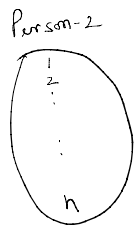
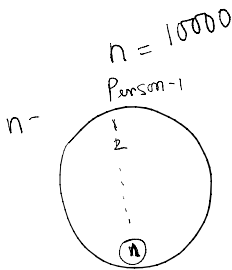
~~$X_n \xrightarrow{P} X$~~

$$\lim_{n \rightarrow \infty} P(|X_n - c| < \epsilon) = 1$$

$\epsilon = 10^{-10}$
 $\epsilon = 10^{-100}$

$n = 1000$ $P(|X_n - c| < \epsilon) = 0.985$

$P(|X_n - c| < \epsilon) = 0.992$



0.9998...

X

$X_1 = 1$
 $X_2 = 6$
 $X_3 = 4$

$1 - \frac{1}{6}$
 $2 - \frac{1}{6}$

$$\frac{1}{6} + \frac{2}{6} + \frac{3}{6} + \frac{4}{6} + \frac{5}{6} + \frac{6}{6} = 3.5$$

$$\lim_{x \rightarrow c} f(x) = 12.$$


Convergence almost surely

$$\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \quad \forall \omega \in N \subseteq \Omega$$

$$P(N^c) = 0$$

$$\Omega = N \cup N^c$$

$$P(\Omega) = 1$$

$$P(N) = 1$$

Weak law of large numbers:

$X_1, X_2, X_3, \dots$ sequence of rvs are independent and identically distributed (i.i.d.), $E(X_n) = \mu \quad \forall n$

$$\text{then } \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu$$

Y_1, Y_2 have identical distribution

$P(Y_1=0) = 0.2$	$P(Y_2=0) = 0.2$
$P(Y_1=1) = 0.8$	$P(Y_2=1) = 0.8$

$$E\left(\frac{(a+bX)^t}{t}\right)$$

$$g(X) = X^2$$

$$E\left(\frac{(a+bX^2)^t}{t}\right)$$

$$P(X=x)$$

$$= \sum_x \frac{a^t + b^t x^{2t}}{t} P(X=x)$$

$g_1(x) = \frac{a^t + b^t x^{2t}}{t}$

$$E(g_1(x))$$

$$\cancel{P} \quad X, Y$$

$$\frac{g_2(x)}{y(x)} = \frac{P(X \leq x)}{\frac{dG(x)}{dx}}$$

$$\cancel{P} \quad \frac{H(y)}{h(y)} = \frac{P(Y \leq y)}{\frac{dH(y)}{dy}}$$